

Enhancement of scale factor accuracy of modulation algorithm for a closed-loop fiber-optic gyroscope.

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June 1, 2026

Abstract

TODO abstract

1 Motivation

An iFOG - Interferometric Fibre Optic Gyroscope - concept is based on the Sagnac effect [1]. Over the years, its design has already been exercised multiple times. However, the limits which hold the development back are pushed forward as technology evolves, enabling for better precision, miniaturized size or lower power consumption. At the same time, the ever progressing industry finds new and more strict applications for the gyroscopes, setting the requirements bar even higher. Such context makes the already existing designs being tested and optimized in every possible aspect. This applies equally to all the optical and electro-optic components, the electronics and the driving software. A crucial part of the control system of an iFOG is the modulator. In a closed-loop design it is a vital element of the feedback loop. It directly affects the overall performance. It drives the optical circuit into a predefined state, in which the circuit's response can be demodulated and error signals determined. Modulator accuracy has critical impact on the resulting performance. This paper takes a closer look at the all-digital implementation of the modulator subsystem and particularly on the algorithm controlling it. Mathematical analysis is conducted to allow for pure-software optimizations in the algorithm's accuracy.

2 Principle of operation - open-loop configuration

A fibre-optic gyroscope utilizes Sagnac effect. A light source generates a light beam, which is ideally split in 50/50 ratio. Each half then enters a loop of relatively long fiber through each of its ends. The fiber length L determines light transit time τ through it - Equation 1, where n is the refraction coefficient, affecting light speed in the medium.

$$\tau = \frac{L}{n \cdot c} \quad (1)$$

The halves go past each other and exit on the opposite ends of the fiber. The optical circuit is arranged, so that the two exiting beams interfere with each other and the resulting light intensity is measured at a detector. When the circuit stays at rest, the beams interfere with zero phase shift ($\Delta\phi = 0$), resulting in maximum intensity P_0 measured at the detector. However, when the loop rotates, the two halves travel paths of a slightly different length, resulting in non-zero phase shift. The relative phase shift at the interference point is rotation dependent and in this paper it is referred to as the Sagnac-induced phase shift

$$\Delta\phi_s = \frac{2\pi \cdot L \cdot D \cdot n}{\lambda \cdot c} \cdot \omega \quad (2)$$

where D - fiber loop diameter, ω - angular velocity, λ - wavelength.

The whole phenomenon has been explained in Figure 1. Both halves enter the loop at the *Entry point*. After the transit time they exit through the same point. However, because of the non-zero rotation, this Entry/Exit point moves, effectively shrinking one path and extending the other. The path difference is relatively small, thus it is expressed as a phase difference in wavelength domain.

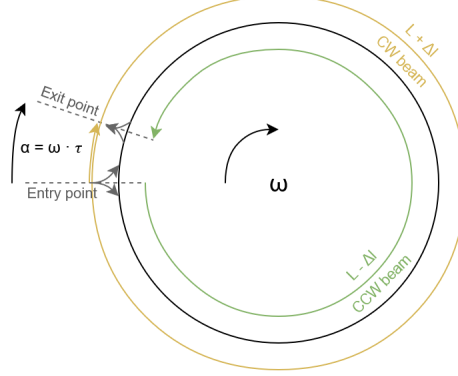


Figure 1: Two counterpropagating light beams enter the fiber loop at the *Entry point*. Their phase is initially in sync. After transit time τ they exit and interfere. During that time the *Entry/Exit point* moves according to the rotation rate ω (by the angle $\alpha = \omega \cdot \tau$). As a result the CW and CCW beams cover non-equal path lengths ($L \pm \Delta l$), which leads to a non-zero phase difference.

The formula in Equation 3, also presented in Figure 2, is the optical circuit response function. It describes how the measured light intensity depends on the total phase shift of interfering light.

$$P(\Delta\phi) = \frac{P_0}{2} \cdot (1 + \cos(\Delta\phi)) \quad (3)$$

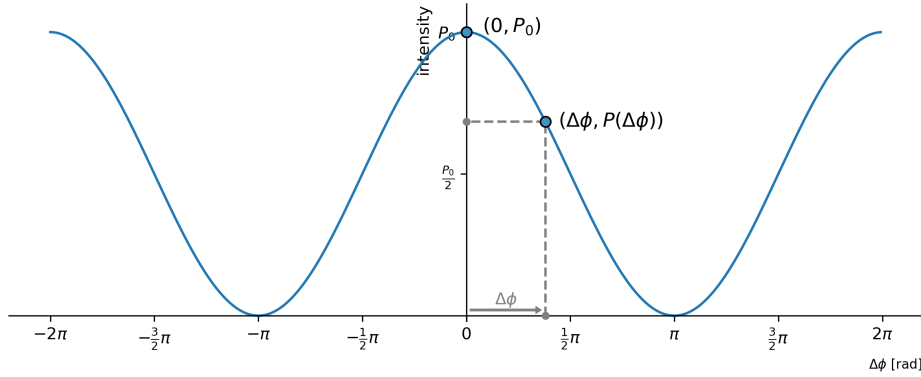


Figure 2: Optical circuit response is a shifted cosine. The two exemplary operating points refer to the circuit being at rest, yielding the maximum possible intensity - $(0, P_0)$ - and subjected to rotation, at non-zero phase shift and decreased intensity - $(\Delta\phi, P(\Delta\phi))$.

When the circuit is at rest ($\omega = 0 \rightarrow \Delta\phi = 0$), the operating point resides at peak intensity P_0 . A non-zero rotation results in phase shift moving the operating point according to rotation rate and direction, decreasing measured intensity. Two problems are linked with this *open-loop* operation. Firstly, intensity drop alone does not allow for determining rotation direction - response function is symmetrical. Secondly, in the resting region, the slope is flat, thus the sensitivity is zero. With greater rotation rate, the slope becomes more steep, reaching its maximum sensitivity in $\Delta\phi = \pm\frac{\pi}{2}$, but then decreases again. For these reasons a *closed-loop* configuration is superior in performance.

3 Modulation scheme - closed-loop configuration

In a closed-loop configuration control system biases the optical circuit sequentially in four predefined points $\Delta\phi \in \{+\frac{3}{2}\pi, -\frac{1}{2}\pi, -\frac{3}{2}\pi, +\frac{1}{2}\pi\}$, utilizing the *four-step* modulation (Figure 3). This allows for linearizing the response, keeping the circuit at maximum sensitivity and detecting rotation direction.

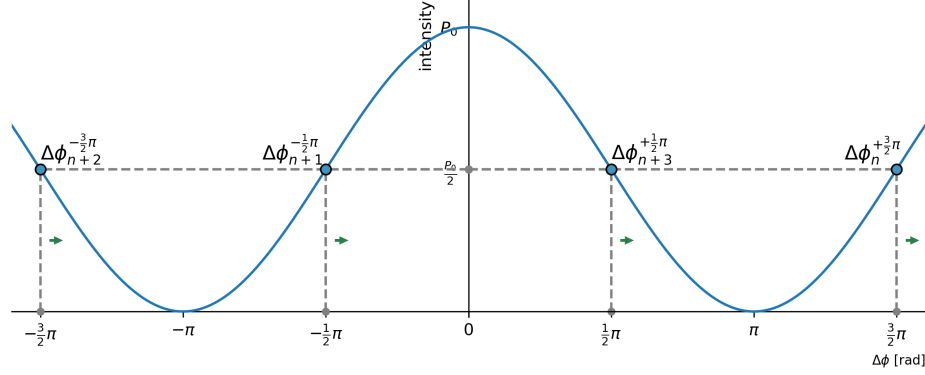


Figure 3: Circuit biased in four points of highest sensitivity.

Compared to *two-step* modulation this scheme is supreme, as it allows for controlling the *scale factor* - an essential parameter for maintaining device performance. Modulator injects the mentioned phase shifts by using an electro-optic actuator placed on the fiber path. The amount of applied shift is controlled by an electronic circuit. It performs digital calculations and generates corresponding voltages via a digital-to-analog converter and an electronic front-end. The voltage is translated to a phase shift according to the half-wave voltage V_π - a parameter of the actuator. Each element in this pipeline is a subject to temperature drift and aging, effects which influence the total combined gain. For this reason, continuous calibration is required for maintaining measurement accuracy and it is realized by tracking the scale factor. This allows the control algorithm to know where the desired bias points are in the digital domain, while they constantly drift.

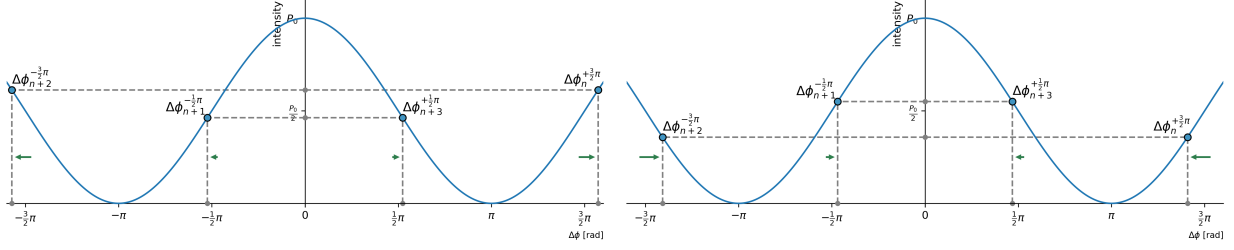
Diagrams in Figure 4 depict how the biased circuit responds to various conditions - misaligned scale factor and uncompensated rotation present. While this paper does not discuss in details how the rotation is measured, it is crucial to emphasize, that different feedback signals can be derived for scale factor adjustment and rotation measurement. These errors allow the digital control loop for adjusting runtime parameters and placing the operating points exactly where they are intended to be.

From control point of view and for simplicity, the combined scale factor can be referred to as the half-wave voltage V_π . Even though the actuator gain drift is only one of the scale factor components, the assumed linearity of others, makes it convenient to aggregate all into a single parameter. Other, non-linear errors, such as DAC INL are out of scope of this analysis. Control system biases the circuit continuously with the following bias point order

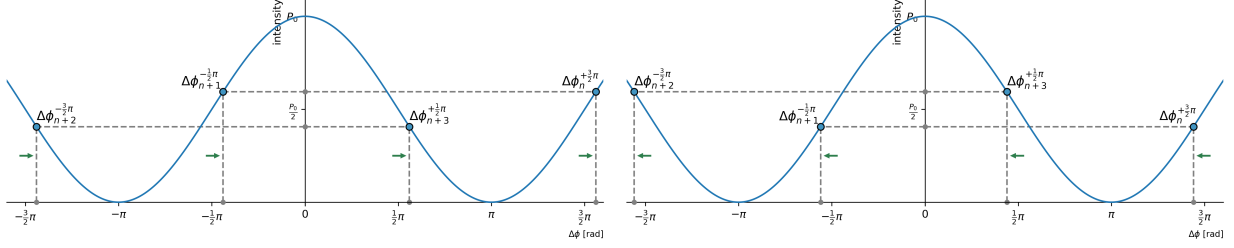
$$\begin{aligned} \{\Delta\phi_{4n+0}^{+\frac{3}{2}\pi} &\rightarrow (+\frac{3}{4}\pi), \\ \Delta\phi_{4n+1}^{-\frac{1}{2}\pi} &\rightarrow (-\frac{1}{4}\pi), \\ \Delta\phi_{4n+2}^{-\frac{3}{2}\pi} &\rightarrow (-\frac{3}{4}\pi), \\ \Delta\phi_{4n+3}^{+\frac{1}{2}\pi} &\rightarrow (+\frac{1}{4}\pi)\} \end{aligned} \quad (4)$$

where the superscripted phase is the intended effective phase shift for the given sample.

Modulator must run with the period of half the transit time $\frac{\tau}{2}$. This allows for the effective phase shift to be a result of a relative difference between every k 'th and $(k-2)$ 'th samples. For this reason the phase



(a) Scale factor too high - bias points are shifted away from zero, resulting in intensity imbalance. (b) Scale factor too low - bias points shifted towards zero, resulting in opposite imbalance.



(c) Phase shift induced by positive rotation - bias points shifted right. Points $\{4n, 4n + 1\}$ correspond to higher intensity than points $\{4n + 2, 4n + 3\}$. (d) Phase shift induced by negative rotation - bias points shifted left. Points $\{4n, 4n + 1\}$ correspond to lower intensity than points $\{4n + 2, 4n + 3\}$.

Figure 4: Four-step modulation allows for detection of scale factor error and Sagnac-induced phase shift.

shift applied by the modulator for each sample can be half of the desired amplitude. Additionally, to null the effect of the rotation-induced (Sagnac-induced) phase shift, modulator adds an ever-updating ramp signal, utilizing Serrodyne modulation with the increment value of $\Delta\phi_r = -\Delta\phi_s$ (equal amplitude, but opposite sign). This way the algorithm nulls out the Sagnac-induced shift, while knowing how much of it it has cancelled, thus knowing the exact rotation rate. In every cycle, the software monitors feedback signals and adjusts control accordingly.

4 Modulator hardware architecture

This paper opts in for an all-digital modulator architecture, while three distinct approaches can be distinguished:

- fully-analog,
- digital with complex analog front end,
- all-digital.

Firstly - a fully analog variant, while feasible, does not comply with today's standards, where information is usually processed in digital form. For integration with external systems, an analog-to-digital interface needs to be there anyway. In such case, incorporating an MCU or DSP in the processing loop brings the advantage of performing certain operations on pure data, rather than on signals carrying these data. In such context, a modern solution shall consist of a digital part for performing mathematical operations and several electronic elements for interfacing with the optical circuit - this resembles a typically implemented solution, with DAC converter and a dedicated gain stage. Moreover, in a world of *software-defined* solutions, an architecture with the minimal amount of electronic components can also be derived. The two latter concepts are described in this section. The all-analog architecture will not be discussed.

4.1 Dedicated gain stage

A reasonable tradeoff between analog and digital approaches is the architecture proposed in . It is composed of a digital part, DAC and dedicated gain stage. The configuration has been depicted in Figure 5.

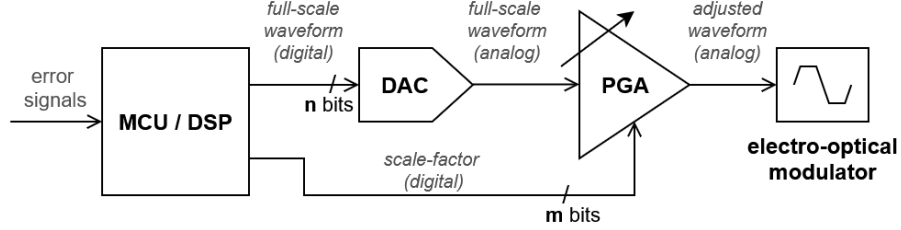


Figure 5: Modulator with a dedicated hardware gain stage. MCU/DSP outputs a fixed-scale digital waveform, which directly drives a DAC. Then, before reaching the electro-optic actuator, scale factor of the analog waveform is adjusted with a programmable gain amplifier (PGA), or a custom gain stage, resulting in adjusted-scale waveform.

In this configuration DAC and PGA are driven with n - and m -bit words. The digital nature of controlling gain stage introduces quantization noise into the control loop. However, one of the important advantages of this setup is the intrinsic separation of the modulation problem into two smaller parts - generating the waveform in fixed-scale and applying the scale factor. The two can be implemented by software without dependence on each other. On the contrary, the dedicated gain stage is a subject to interferences. It may have its own nonlinearities and introduce delay in the signal path. Most importantly, its bandwidth may be gain dependent, distorting the waveform.

4.2 All-digital approach - software-defined solution

The all-digital approach depicted in Figure 6 utilizes the availability and wide range of high precision digital-to-analog converters. It allows for disposing the physical gain stage, by applying scale factor digitally, prior to driving the DAC. This requires DAC reference voltage to be high enough to cover all possible ranges. This approach also introduces quantization noise. These two disadvantages, are similarly the problems of the previous solution. The main advantage this idea brings is the minimization of the amount of electronic components used. This yields less sources of potential nonlinearities and electromagnetic interference. Since the electro-optic modulator has high input impedance, the output of a voltage-DAC can be directly connected to it, omitting any buffers on the signal path. The profit resulting from less noisy electronics copensate the unavoidable DAC resolution loss - its full range may never be utilized because of the required margins. Moreover, the following sections discusses an algorithmic approach, which regains one bit of accuracy, when using the all-digital architecture.

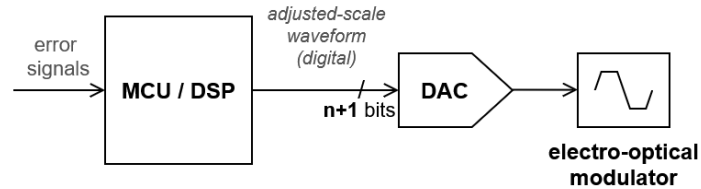


Figure 6: .

Additional advantage of leaning towards the proposed architecture is, that the whole design becomes a software-defined solution. As in other fields, it brings flexibility from the R&D process perspective. While iterating hardware design is a time-consuming process, updating software is relatively quick and cos effective. Offloating the whole problem to the software side, obviously makes teh implementation more complex, but it limits the amount of interaction at the boundary of software and hardware development. From this

perspective the net workload remains at the same order of magnitude, offering slightly different, modern, approach.

4.2.1 Voltage reference

For an all-digital modulator an adequate reference voltage has to be set up to account for the maximum possible scale factor. From Equation 4 it results, that a span of 1.5π is required for biasing and additionally, the Serrodyne modulator requires a span of 2π . This yields the required span of $S = 3.5\pi$. Minimum DAC reference V_{ref}^{min} shall account for it with some margin $M \in [1.0; 1.1]$.

$$V_{ref}^{min} = V_{\pi}^{max} \cdot S \cdot M \quad (5)$$

The V_{π}^{max} denotes the highest expected half-wave voltage. When determining final reference voltage it is advantageous to round it up to a multiple of one of typically available chip references (Equation 6), e.g. $K \cdot 5V$ or $K \cdot 4.096V$. It makes implementing the electronic circuit more straightforward. In practice, a fair compromise is to utilize the resulting voltage excess for the margin M , so that after adding it, the $V_{ref} \approx V_{ref}^{min}$.

$$V_{ref} = \lceil V_{ref}^{min} \rceil \quad (6)$$

4.2.2 Effect of the quantization error

Usage of a DAC results in quantization error. Converter resolution and its reference voltage determine the minimal controllable voltage change as

$$\Delta v = V_{ref} \cdot 2^{-N} \quad (7)$$

In phase domain the scale factor can be controlled with the accuracy corresponding to Δv relative to scale factor

$$\Delta \phi_v = \frac{\Delta v}{V_{\pi}} \quad (8)$$

It can vary over time, but rather slowly in the context of modulation frequency. The resulting equation depends of a few parameters, while quantization error is half of its value.

$$\Delta \phi_v = \frac{\lceil V_{\pi}^{max} \cdot S \cdot M \rceil \cdot 2^{-N}}{V_{\pi}} \quad (9)$$

$$\Delta \phi_q = \frac{1}{2} \cdot \Delta \phi_v \quad (10)$$

Exemplary calculation, based on real-world values yields

$$\begin{aligned} \Delta \phi_v^{ex} &= \frac{\overbrace{[4.5V]}^{V_{\pi}^{max}} \cdot \overbrace{3.5}^S \cdot \overbrace{1.025}^M \cdot \overbrace{2^{-16}}^{2^{-N}}}{\underbrace{3.5V}_{V_{\pi}}} = \frac{[16.14V] \cdot 2^{-16}}{3.5V} = \frac{\overbrace{16.384V}^{4 \cdot 4.096V} \cdot 2^{-16}}{3.5V} = \\ &= 2^{-16} \cdot 4.68 = 2^{-16} \cdot 2^{2.23} \\ \Delta \phi_q^{ex} &= \frac{1}{2} \cdot \Delta \phi_v^{ex} = 2^{-16} \cdot 2^{1.23} \end{aligned} \quad (11)$$

From this it is clear, that the quantization error, bound to controllable resolution can be improved by selecting a converter with greater number of bits. However, the resolution loss will always stay at the level of over two bits. It must be noted, that most of this loss is due to the span S , required for proper operation of the algorithm. It is not a consequence of using the proposed all-digital architecture.

In each of the bias points, the following drive levels are possible.

$$\begin{aligned}
\phi_{4n+0}^{+\frac{3}{2}\pi} &= k_{4n+0} \cdot \Delta\phi_v = +\frac{3}{4}\pi + \delta_{4n+0} \\
\phi_{4n+0}^{-\frac{1}{2}\pi} &= k_{4n+1} \cdot \Delta\phi_v = -\frac{1}{4}\pi - \delta_{4n+1} \\
\phi_{4n+0}^{-\frac{3}{2}\pi} &= k_{4n+2} \cdot \Delta\phi_v = -\frac{3}{4}\pi - \delta_{4n+2} \\
\phi_{4n+0}^{+\frac{1}{2}\pi} &= k_{4n+3} \cdot \Delta\phi_v = +\frac{1}{4}\pi + \delta_{4n+3}
\end{aligned} \tag{12}$$

Value $k \in [-2^{N-1}, 2^{N-1} - 1]$ is an integer, in range of the used DAC (of N resolution). By incrementing/decrementing k_n the error δ_n can be controlled with the resolution of $\Delta\phi_v$. However, because of the quantization error, its value can always be up to $\Delta\phi_q$.

5 Precision estimation and impact

To close the control loop, scale factor error feedback is required. It is calculated from the incident light intensity difference between two consecutive bias points in each pair. The total phase shift difference $\Delta\phi_k$, affecting intensity for the sample k , results from the current drive ϕ_k and the drive two points prior.

$$\Delta\phi_n = (\phi_n - \phi_{n-2}) + \underbrace{\Delta\phi_{s,n} + \Delta\phi_{r,n}}_{=0} \tag{13}$$

The values of $\Delta\phi_{s,n}$ and $\Delta\phi_{r,n}$ denote the Sagnac-induced phase shift and the nulling ramp shift, which are assumed to be zero for this analysis. By using equations 3 and 4 and assuming the $\cos$ function linear near the bias points (Equation 15), intensity values at each bias point are

$$\begin{aligned}
P_{4n+0}^{+\frac{3}{2}\pi} &= \frac{P_0}{2} \cdot (1 + \cos(\Delta\phi_{4n+0}^{+\frac{3}{2}\pi})) = \frac{P_0}{2} \cdot (1 + \cos(\phi_{4n+0}^{+\frac{3}{2}\pi} - \phi_{4n-2}^{-\frac{3}{2}\pi})) = \\
&= \frac{P_0}{2} \cdot (1 + \cos((\frac{3}{4}\pi + \delta_{4n+0}) - (-\frac{3}{4}\pi + \delta_{4n-2}))) = \\
&= \frac{P_0}{2} \cdot (1 + \cos(\frac{3}{2}\pi + \delta_{4n+0} + \delta_{4n-2})) = \\
&= \frac{P_0}{2} \cdot (1 + (\delta_{4n+0} + \delta_{4n-2})) \\
P_{4n+1}^{-\frac{1}{2}\pi} &= \frac{P_0}{2} \cdot (1 - (\delta_{4n+1} + \delta_{4n-1})) \\
P_{4n+2}^{-\frac{3}{2}\pi} &= \frac{P_0}{2} \cdot (1 + (\delta_{4n+2} + \delta_{4n+0})) \\
P_{4n+3}^{+\frac{1}{2}\pi} &= \frac{P_0}{2} \cdot (1 - (\delta_{4n+3} + \delta_{4n+1}))
\end{aligned} \tag{14}$$

$$\cos(\phi_b + \Delta\phi) = \underbrace{\cos(\phi_b)}_{=0} + \cos'(\phi_b) \cdot \Delta\phi = -\sin(\phi_b) \cdot \Delta\phi \quad \text{for } \phi_b \in \{+\frac{3}{2}\pi, +\frac{1}{2}\pi, -\frac{1}{2}\pi, -\frac{3}{2}\pi\} \tag{15}$$

Feedback signal can be calculated at samples $4n+1$ and $4n+3$ and yields

$$\begin{aligned}
e_{4n+1}^{-\frac{1}{2}\pi} &= P_{4n+1}^{-\frac{1}{2}\pi} - P_{4n+0}^{+\frac{3}{2}\pi} = -\frac{P_0}{2} \cdot (\delta_{4n+1} + \delta_{4n+0} + \delta_{4n-1} + \delta_{4n-2}) \\
e_{4n+3}^{+\frac{1}{2}\pi} &= P_{4n+3}^{+\frac{1}{2}\pi} - P_{4n+2}^{-\frac{3}{2}\pi} = -\frac{P_0}{2} \cdot (\delta_{4n+3} + \delta_{4n+2} + \delta_{4n+1} + \delta_{4n+0})
\end{aligned} \tag{16}$$

It can be generalized for every odd bias point, where it shall be calculated - this results from the demodulation algorithm assumptions.

$$e_{2n+1} = -\frac{P_0}{2} \cdot (\underbrace{\delta_{2n+1}}_{\text{current}} + \underbrace{\delta_{2n+0} + \delta_{2n-1} + \delta_{2n-2}}_{\text{three prevoius errors}}) \quad (17)$$

The error signal, used for control loop closure, is dependent on four recent drive values. Different weights are assigned to even and odd samples. Because the DAC can be controlled with Δv resolution, the scale factor error, and thus the occuring errors δ_n are all controllable with some minimal phase difference $\Delta\phi^{min}$ corresponding to Δv .

$$\Delta\phi^{min} \equiv \Delta v \quad (18)$$

6 Algorithm design